# How Forms Become Reconstructible

## Acquired Records, Reciprocal Repair, and Functional Renewal

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## Abstract

How can a pattern or learned response be rebuilt after damage or replacement? Using computer models, we distinguish what past experience leaves behind, what a system does now, and what a replacement can recover. We call the information carried forward a record, and test how systems use it and write a new one.

In spatial models, records of patterns formed without a supplied target image help later systems recover features specific to the original pattern. This modest effect persists through sixteen replacements, even as measured reconstruction becomes less accurate. Surviving organization in a damaged system can improve a record's usefulness to a new system. Changing how that record is read or rewritten also changes what is rebuilt.

In a separate network model, one of two simple taught response rules can be recovered after 32 damaged rebuilds, despite failures along the way. Different rebuilding conditions make success depend on different features of the record. A repair procedure fixed before testing improves later reconstruction and, in a separate test, helps more systems keep working throughout repeated rebuilding. In a model of communicating cells, cells sharing the same broad regulatory state label differ in their ability to pass that state to replacements. Changing specific internal components or communication timing changes this ability.

Rebuilding can restore function after specified losses, but preserving a record unchanged often works better. Some successes remain sensitive to numerical details, and better forecasts do not always yield better repair choices. The central finding is that retaining information, expressing it now, and using it to rebuild later are different capacities. The models provide controlled ways to study these differences; a general predictive theory, inherited flexible problem-solving and validation in living systems remain open.

**Keywords:** reconstructibility; acquired memory; reciprocal repair; functional renewal; embodiment; plastic heredity; bounded observer.

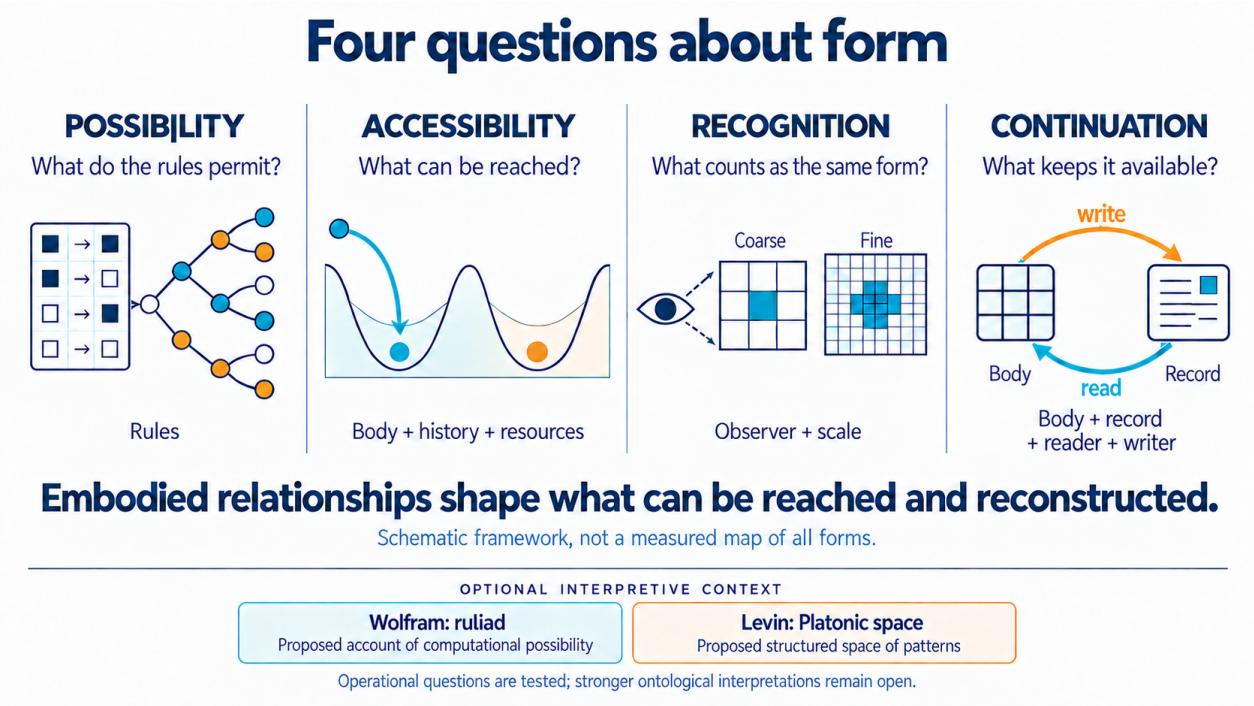



**Figure 1: Four questions about form.** Possibility, embodied access, observer-dependent recognition and continuation are distinct explanatory questions. Reading and writing act within the dynamics; the observer reports a distinction. The landscape and body–record loop are conceptual schematics, not a measured energy surface or a universal architecture. These simulations test access, recognition and reconstruction, questions motivated by the ruliad and Levin's proposed Platonic space. Their results do not distinguish the stronger ontological interpretations from alternative accounts.

## 1. From an occurrence to a capacity to return

A pattern can reappear because a system repeatedly converges on the same outcome. It can also reappear because a particular past episode left information that changes what develops next. These explanations become distinguishable when we change the history, remove the original body and test what a successor reconstructs. Here, a form is an organization or response pattern identified by declared measurements.

**Our contribution is to separate acquired historical specificity, current expression and successor usefulness across controlled reconstruction boundaries, then identify embodied relationships and record features that change successor usefulness.** The models answer complementary questions; they do not establish one mechanism operating across every substrate.

This contribution builds on established accounts of pattern formation and memory. Turing showed how local dynamics can generate organization without a picture of the final form. Hopfield networks reconstruct patterns from incomplete cues; epigenetic models restore marks through noise and turnover; neural cellular automata distinguish growth, maintenance and repair. Turing, 1952; Hopfield, 1982; Dodd et al., 2007; Mordvintsev et al., 2020

Two closer comparisons help locate the experimental addition. Ryan and colleagues recovered memory by directly activating engram cells under induced amnesia, associated with retained engram-specific connectivity. Our question extends from availability despite failed retrieval to what a retained and renewed record lets a replacement reconstruct. Owen, Osmanović and Mirny model how chromatin contacts, reader–writer enzymes and limited enzyme abundance support memory under erosion. Our acquired-history manipulations, fresh-body boundaries and controlled record and repair interventions test related dependencies under different supplied mechanisms. Ryan et al., 2015; Owen et al., 2023

The companion *Plastic Heredity* studies episodes in which parent–descendant resemblance breaks and returns, together with rulebook-conditioned access and engineered carriers. Renewal of that measured relation does not itself establish renewal of an acquired information carrier. This paper follows the latter problem through five questions: what crosses replacement, what embodied relationships affect its usefulness, which record features support reconstruction, whether those features improve prediction or repair, and what a general theory would still have to predict. Bjarnason, 2026, companion v5

## 2. Models, boundaries and measurements

### 2.1. What is retained and what is replaced?

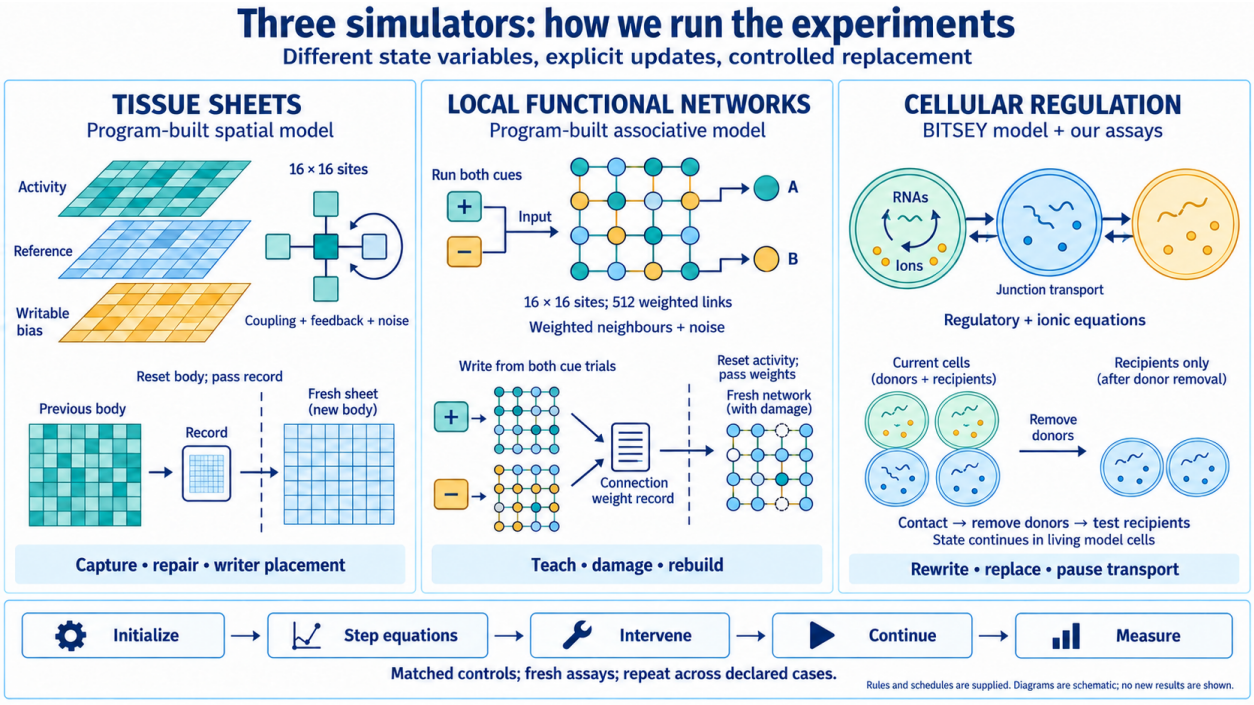



**Figure 2: Three simulators and their computational boundaries.** Spatial records and connection weights enter fresh activity states; cellular assays transmit history through live simulated cells. The two-cue writer and donor-removal boundaries are explicit. Schematic patches, not measured outputs. Record roles vary by assay; SI methods C.1–C.44 specify native equations and protocols.

| Main setup | Origin and retained variables | Experimental role |
|---|---|---|
| **Continuous tissue sheets** | Our dimensionless 16 × 16 periodic sheets: signed activity V, internal reference R and writable threshold field theta. | Spatial history, reciprocal repair and writer placement. |
| **Local functional networks** | Our spatial activity model with 512 reciprocal neighbor connections; experience writes their weights. | A taught two-rule response through fresh activity states and repeated damage. |
| **Cellular regulatory model** | The published BITSEY regulatory/ionic model and junction transport, with added transmission assays. | Editable regulatory state through changing cellular membership. |

The first two are mechanism models constructed within this program. BITSEY supplies a published biological modeling context; our extensions retain native regulatory, ionic and transport laws while adding cues, contacts, replacement and interventions. A separate Hopfield comparison tests reader conditions in an associative-memory architecture. Earlier GARD, cellular-automaton and Wagner-style models supply supporting continuity and carrier comparisons, rather than additional panels of the same experiment. Grodstein et al., 2026; Segré et al., 2000; Wolfram, 1984; Wagner, 1996

The **rulebook** supplies dynamics and constraints. The **body** is their currently instantiated organization. A **record** retains state from an earlier episode; a **reader** lets that state affect the body, and a **writer** changes or reconstructs it. The **observer** defines the measured distinction. Changing an observer can change reported continuity without changing a trajectory; changing a reader can change the trajectory. A measurement fed back into control belongs to the causal apparatus.

The common experimental route is **history → record → developing body → successor record → fresh embodiment**. Only the declared record crosses a packet-replacement boundary. During body-driven renewal, the incoming record is unavailable as a direct writing input. In reciprocal repair, we intervene on surviving organization and test only its resulting record elsewhere. Shared guides, task cues and recipient machinery remain part of the apparatus.

Spatial records generally allocate 2048 bytes to 256 float64 values; the compact branch allocates 128 bytes to sixteen values. Functional records allocate 4096 bytes to weights. These are storage budgets, not measured information contents. Cellular transmission has a different boundary: complete states continue within live simulated cells as new recipients replace previous carriers; no cell division is simulated. The Supporting Information (SI), methods C.1–C.44, retains equations, protocols and campaign identities.

## 2.2. Conditional reconstructibility

For an assay with a binary expression criterion, define

$$C_H^O(r, m \mid \mathcal{A}, \mathcal{D}) = \Pr[O(B_H, m) = 1 \mid r, \mathcal{A}, \mathcal{D}] .$$

Here $r$ is the initial record, $B_H$ the fresh-descendant assay after $H$ completed record renewals, and $m$ the evaluated form or response; $H = 0$ means reading before renewal. The apparatus $\mathcal{A}$ specifies body dynamics, reader, writer, shared guides, supplied inputs and time budgets. $\mathcal{D}$ specifies births,

damage, noise and their sampling; observer $O$ defines successful expression and its thresholds. The evaluated form need not have been supplied as a developmental target.

This defines a quantity to estimate, not a predictive law. Repeated continuations from an exactly restored record under the same declared future distribution could estimate its conditional capacity. Cohort averages additionally mix records and histories. Historical specificity is a separate matched own-versus-foreign-record contrast, not the total probability of success. High terminal capacity also differs from uninterrupted expression throughout a lineage.

**Episodic hereditary continuity**, **inheritance of acquired information** and **carrier renewal** are distinct properties. Resemblance can resume without a newly acquired memory; protected copying can transmit an acquired record without either a resemblance break or body-driven renewal. Likewise, a compact packet can depend on a larger shared decoder. Figure 1's possibility, accessibility, recognition and continuation are explanatory questions, not successive developmental stages. E01; E17

### 2.3. Evidence status at a glance

| Evidence claim | Status | Scope and evidence |
|---|---|---|
| Particular acquired history affects a replacement. | Established | Unprescribed capture and compact renewal retain modest own-history specificity. E31; E43 |
| Body arrangement and writer relationships change successor usefulness. | Established | Reciprocal repair, writer placement and sustained reader conditions have causal effects within their respective models. E09; E10; E29; E86 |
| A taught function remains reconstructible after repeated damage. | Established | Terminal success differs from uninterrupted expression; mechanisms depend on rebuilding noise. E63; E64; E69; E76 |
| A fixed record-based procedure improves reconstruction. | Established | Separate fresh tests support one-read repair and serial maintenance. E78; E83 |
| Record structure improves forecasts beyond expression/history. | Established | Frozen predictions improve on two comparators; the separate decision task ties them. E71 |
| A shared cellular classification specifies transmission readiness. | Rejected within these panels | Component substitutions and communication timing change readiness despite the same coarse label. E65; E85 |
| Numerical-robustness requirements are fully met. | Criteria not met | Joint challenges retain high independent-noise success, but fail the five-point noninferiority requirement. E70; E75 |
| The body supplies message-specific correction, or repeated reference writing reliably preserves the distinction. | Criteria not met | Donor redirection fails; the four-message reference chain loses readable content. E10; E12 |
| Rewriting generally matches copying, or diagnostic challenges improve forecasts beyond passive observation. | Unresolved or condition-dependent | Protected copying often wins; the prospective challenge contrast remains unresolved. E27; E29; E55 |
| Accurate forecasts provide a superior repair policy. | Not established | A warning policy fails its directional prediction; repair efficiency remains unresolved. E88 |

Established means supported within the declared model and test. Failed criteria can coexist with positive effects; unresolved comparisons establish neither equality nor absence. This is one human-directed program: blocks, related histories, repeated readouts and numerical resolutions remain

distinct inference units. Intervals retain their original coverage and adjustment. The evidence selection is unchanged from v1.0; no later result is assumed.

## 3. What acquired distinction survives a replacement?

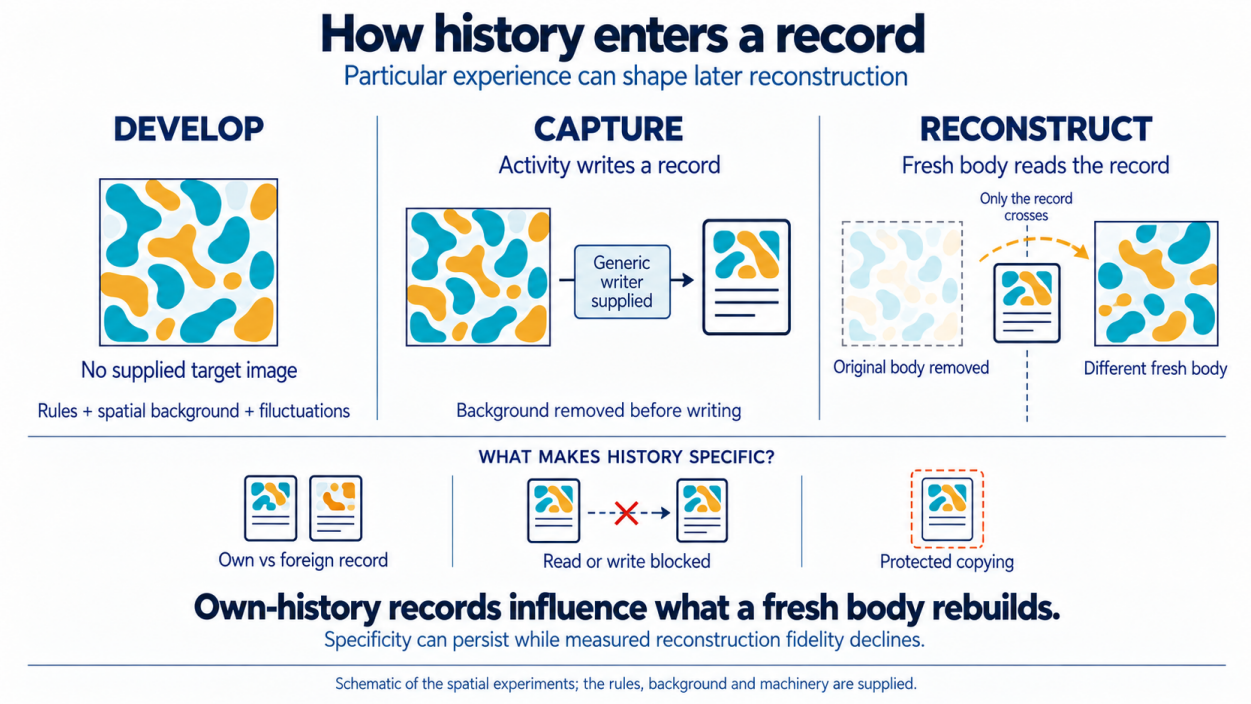



**Figure 3: How history enters a record.** Schematic of unprescribed spatial development, activity-based capture and fresh-body reconstruction. The spatial background and generic writer are supplied; the background is removed before writing, and only the record crosses replacement. Own-versus-foreign records and reading/writing controls test historical specificity; protected copying supplies a preservation comparison. The pictured shapes illustrate the information route, not saved fields or an estimate of fidelity. E31; E43

### 3.1. Unprescribed spatial history

In the **unprescribed-capture experiment** (E004), four founders share each randomly drawn spatial background but have independent initial states and noise. No target image, label or reward is supplied. The background is removed before a generic activity-driven writer captures a record. Only this record reaches descendants; the founder's observed pattern remains available solely for offline evaluation. Own-record and matched foreign-record chains are both scored against the same own-founder prototype. E31

After three read–erase–rewrite replacements and a fresh read, own records exceed foreign records by **1.86 qualified-agreement points [1.22, 2.56]**, a 95% whole-block interval across **32 fresh blocks and 128 founders**. Reading- and writing-disabled controls support the required routes. Protected copying nevertheless exceeds renewal by **5.42 points [4.80, 6.05]**. Particular history therefore matters even when reconstruction preserves it imperfectly. E31

The spatial observer is

$$Q(V,F) = \frac{100}{256}\sum_i \mathbf{1}\{|V_i| > 0.2\}\mathbf{1}\{b(V_i) = b(F_i)\}, \qquad b(x) = \mathbf{1}\{x > 0\}.$$

All 256 sites stay in the denominator. Q combines sign matching with amplitude qualification, so it is not ordinary binary-pattern accuracy. This definition applies to the unprescribed-capture and long-renewal comparisons below.

The **compact long-renewal experiment** (E014) encodes the field as sixteen spatial block means, expanded by a fixed decoder. After sixteen rewrites, own-record agreement is **64.38%**, foreign-record agreement **63.00%**, and protected-copy agreement **68.27%**. The own-history increment is **1.387 points [0.848, 1.954]**, across **64 fresh blocks and all 256 founders**, with a 95% whole-block interval. The larger total agreement is not the amount of founder-specific information retained (Figure 4). Formation, acquisition and route criteria pass, with no successful-founder selection. E43

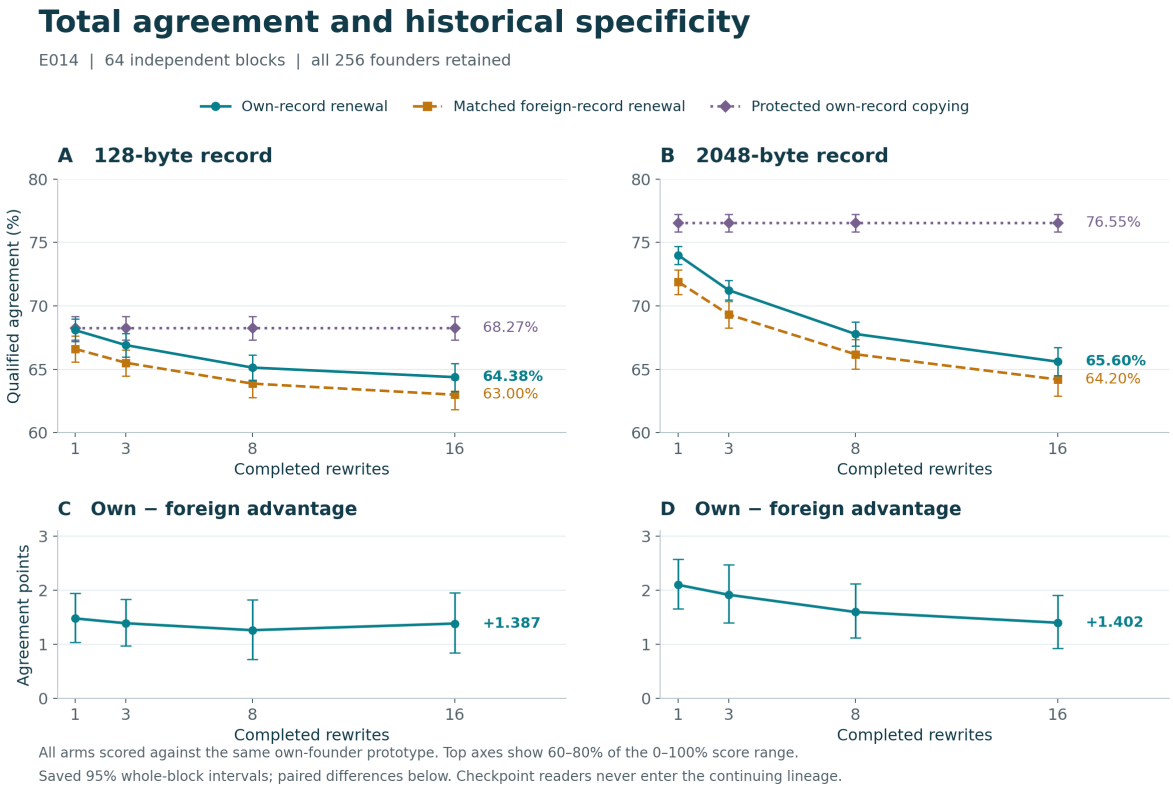



**Figure 4: Total agreement and historical specificity are distinct.** A/B: own-record renewal, matched foreign-record renewal and protected own-record copying, at both declared record budgets and every fresh-reader checkpoint. All arms are scored against the same own-founder prototype; foreign founders share its background. C/D: the paired own-minus-foreign differences and their saved intervals. All 256 founders in 64 independent blocks remain, with two recipient replicates per founder. Bars reproduce native pointwise 95% whole-block intervals; the compact after-16 difference is the primary endpoint. Top axes display 60–80% of the native 0–100% qualified-agreement scale. Checkpoint births and noise are shared across horizons, making protected-copy probes exactly constant; probes never enter the lineage. The figure separates total agreement from historical specificity, but does not separate signs from amplitude qualification. E43

Compact qualified agreement declines from **66.91% at three rewrites to 64.38% at sixteen**. In a post hoc analysis of the same retained fields, sign agreement falls **3.005 points [2.420, 3.630]** while amplitude qualification rises **0.892 points [0.734, 1.046]**. The decline therefore cannot be explained simply by fewer sites crossing the amplitude threshold. These pointwise 95% block intervals describe coordinate agreement, not translation-tolerant shape; SI Figure S5 preserves the full joint-count decomposition. E43

Formation and historical dependence are also separable. A homogeneous-background pilot has a much larger own-history effect, **27.36 points**, but only **13/32 founders** pass formation and the complete sequence fails. The fresh heterogeneous confirmation and this exploratory pilot are different cohorts. Supplied dynamics, geometry and generic writing remain conditions of formation even when a target image is absent. SI C.10/C.16 retain the pilot, budgets and complete controls. E31; E43

### 3.2. Acquiring a reference and revising a cellular lineage

In the **reference-acquisition experiment** (D099), sustained exposure changes the reference subsequently defended. Holding writing or disabling reading removes the body-side effect. The weaker exposure-by-writing contrast after reversing the two target roles is **64.38 percentage points [58.72, 70.09]**, a 95% whole-tissue interval over eight blocks. Reversed recovery nevertheless averages **79.15%**, below its 80% benchmark. Acquisition is supported within the supplied two-pattern family; the full qualification fails. E08

The **editable cellular lineage** (P11) tests a different boundary. Temporary M cues rewrite three E cells; after cue withdrawal and isolated retention, their acquired state passes through three complete handovers. The actual third-round carriers then receive E cues and, after withdrawal, transmit through three further handovers into fresh M recipients. Teachers are removed at each boundary. Both written-M and time-matched ordinary-M origins pass at two numerical resolutions; blocking the RNA route or communication boundary prevents transmission. Figure 5 shows the retained continuous trajectory, including cues, holds and thresholds. E30

This demonstrates editable transmission through actual simulated replacements, with ordinary origins equally successful. E and M are coarse regulatory labels, not complete states or health diagnoses. The same-label comparison becomes consequential in section 7: transmission readiness can change without a different current classification.

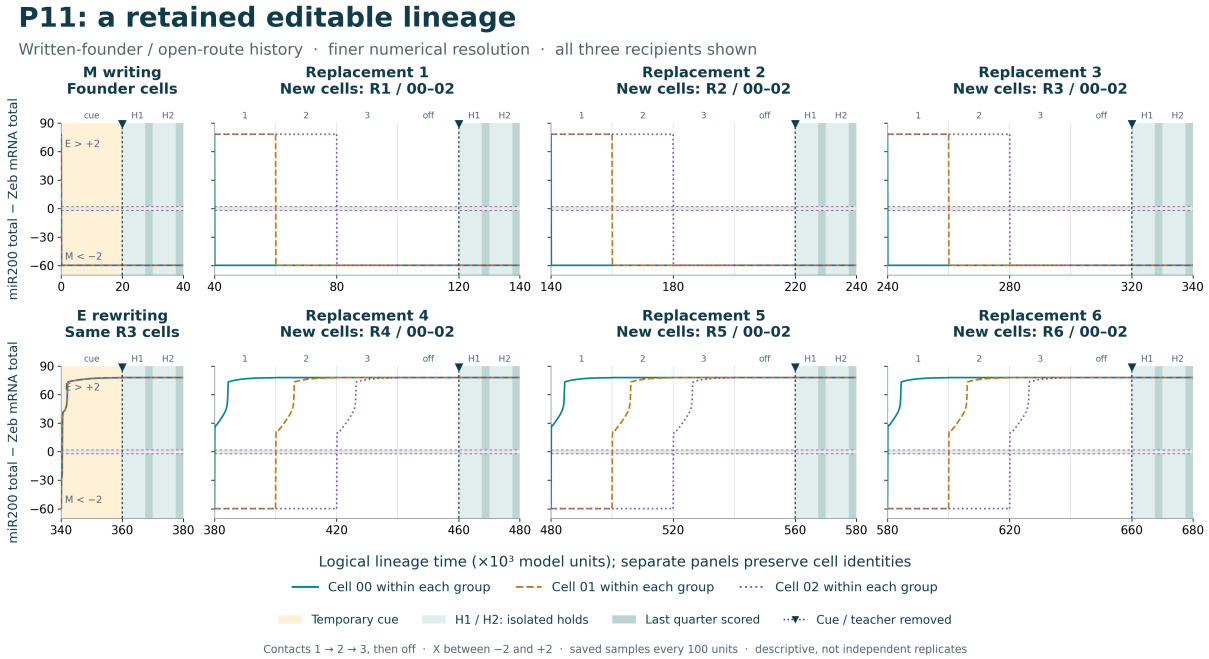



**Figure 5: The retained editable lineage beneath its coarse labels.** The declared written-founder/open-route example at maximum integration step 50. Curves show total miR200 minus total Zeb mRNA at saved 100-unit intervals: E above +2, M below −2, X otherwise. Replacement panels contain new cells; E rewriting acts on the actual third-round carriers. Shading marks cues and successive contacts; markers identify cue or teacher removal. H1/H2 are separate 10,000-unit isolated holds, with scored final quarters darker. Axes use logical model time and one linear activity scale; lines connect samples without smoothing. This identifier-selected illustration was chosen before field inspection but after aggregate results were known. It is one deterministic history, not a population estimate; numerical-resolution pairs are not independent replicates. E30

## 4. How do embodied relationships change later usefulness?

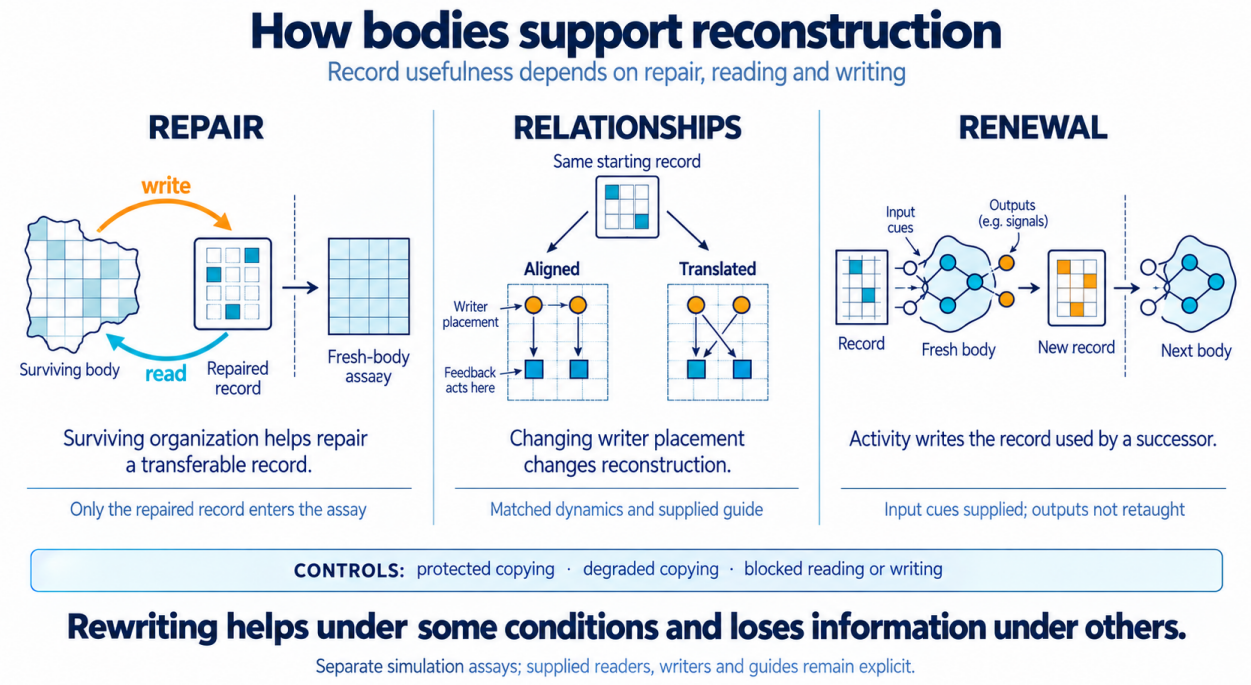



**Figure 6: How bodies support reconstruction.** Three separate assays test surviving organization in record repair, writer placement during reconstruction, and activity-based renewal of a taught function. Only the declared record enters a fresh-body assay or crosses replacement. Copying and blocked-route controls establish different comparisons within their respective apparatuses. These are mechanism schematics; the panels do not describe one combined experiment or guarantee that rewriting improves preservation. E09; E10; E29; E63; E69

### 4.1. Surviving organization repairs a transferable record

The **reciprocal-repair experiment** (D073) compares matched damaged records and half-body fission. Both arms write from recovering activity; one keeps the damaged reader fixed, while the other lets newly written values feed back into recovery. Only the final record enters fresh-body assays. Neither the fragment nor its working reference accompanies it. E09

Live feedback raises recognition from **88.65% to 90.67%**, an increment of **2.03 points [0.85, 3.59]** across sixteen tissue blocks. A separate **arrangement intervention** (D074) preserves or scrambles the surviving body while matching both damaged records and body-value distributions. Preserving arrangement improves later own-message recognition by **1.78 points [1.10, 2.51]** across sixteen further blocks. Each interval has nominal 98.33% coverage within a three-contrast family; fresh readouts are nested within tissues. E09; E10

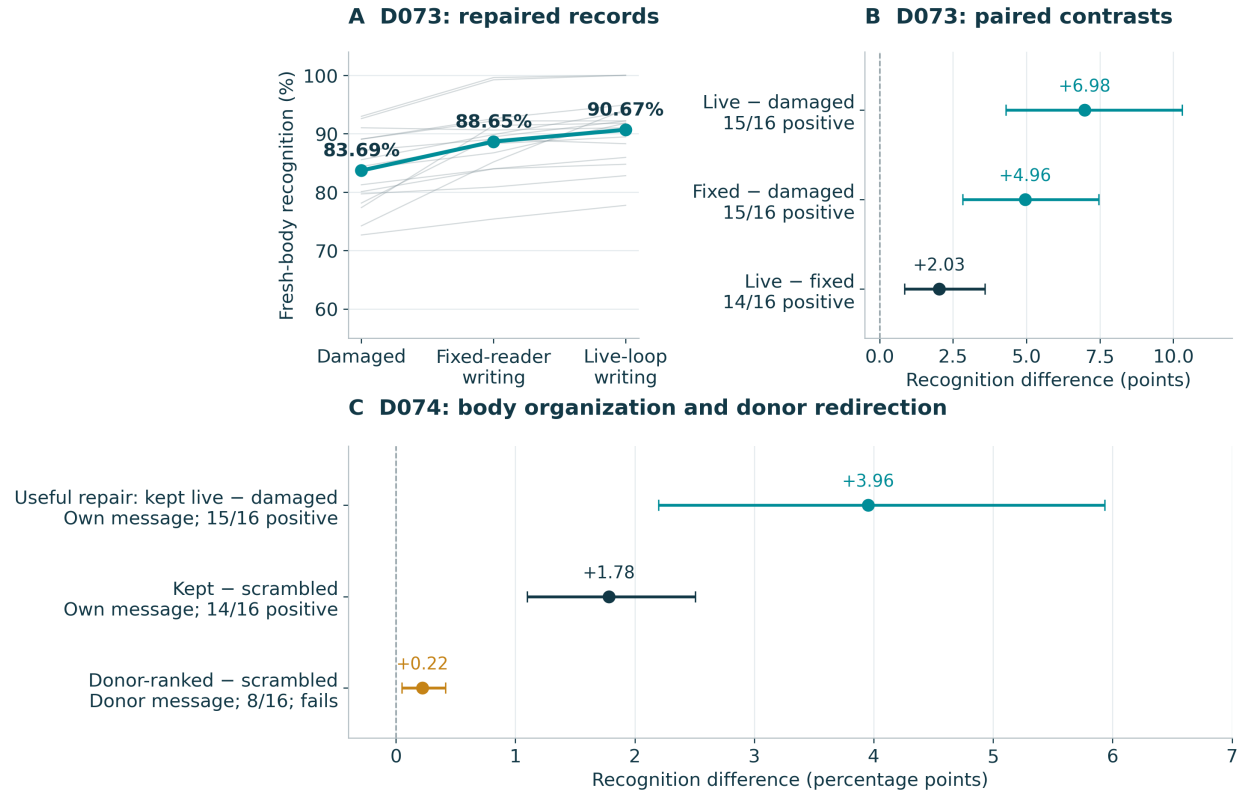



**Figure 7: Surviving organization improves a transferable record.** Top: reciprocal repair on sixteen tissues, showing recognition means, paired tissue traces and the three predefined contrasts. Bottom: spatial-organization interventions on sixteen further tissues, retaining each contrast's own-message or donor-message endpoint. Bars reproduce saved intervals with nominal 98.33% individual coverage; qualification also requires at least twelve positive tissue differences. Donor redirection fails this consistency criterion despite a positive interval. Only repaired theta enters the fresh-body assay. E09; E10

These interventions identify surviving organization as a contributor to record usefulness elsewhere. The stronger donor-redirection criterion fails its required consistency across tissues. Arrangement could supply corrective content, or help exploit content still present in the damaged record; message-specific correction remains unresolved. SI C.1–C.2 specifies the four half-plane messages and the matched controls.

### 4.2. Writer placement and reader history

The **writer-placement test** (D135) and its fresh-founder confirmation (D140) begin paired chains from identical records. A shared guide, fitted from sixteen separate development records, correlates each packet against two supplied family templates across all 256 circular translations. It inverse-translates the incoming field itself; it neither replaces nor blends its values with template pixels. The 4096-byte guide belongs to the apparatus, outside the inherited 2048-byte record. E29

After a fresh body reads for 300 updates, the packet and guide become unavailable as direct writing inputs. Activity, internal reference and a moving average remain. A new theta field starts at zero and is updated for 1000 steps. One arm applies it at corresponding sites; the other uses one uniformly sampled nonidentity circular translation, fixed during that generation. The successor record is derived from the final reference, $\Pi_{0.3}[\tanh(R_{\text{final}})]$, mean-centered and normalized to standard deviation 0.3. Final theta itself is not transmitted. Three rewrites precede a fourth, non-writing read (Figure 8; SI C.8).

The primary endpoint is **qualified preference**:

$$P_q(V) = \frac{100}{256} \sum_i \mathbf{1}\{|V_i| > 0.2\}(u_i - a_i),$$

where u and a indicate sign agreement with the inherited and alternative targets. All sites remain in the denominator; the targets disagree at 128 of 256 sites, so scores range from **−50 to +50 points**. A change can reflect signs, amplitude qualification, or both.

Across **eight new founder blocks**, aligned feedback improves terminal preference by **2.89 points [1.72, 4.17]**, a 95% whole-block interval. A secondary sign-only readout gives **3.11 points [1.92, 4.39]**, with an unadjusted pointwise interval. The direction survives threshold removal, without assigning separate causal shares to signs and amplitudes. The pooled fresh-cohort comparison with protected copying remains unresolved. E29

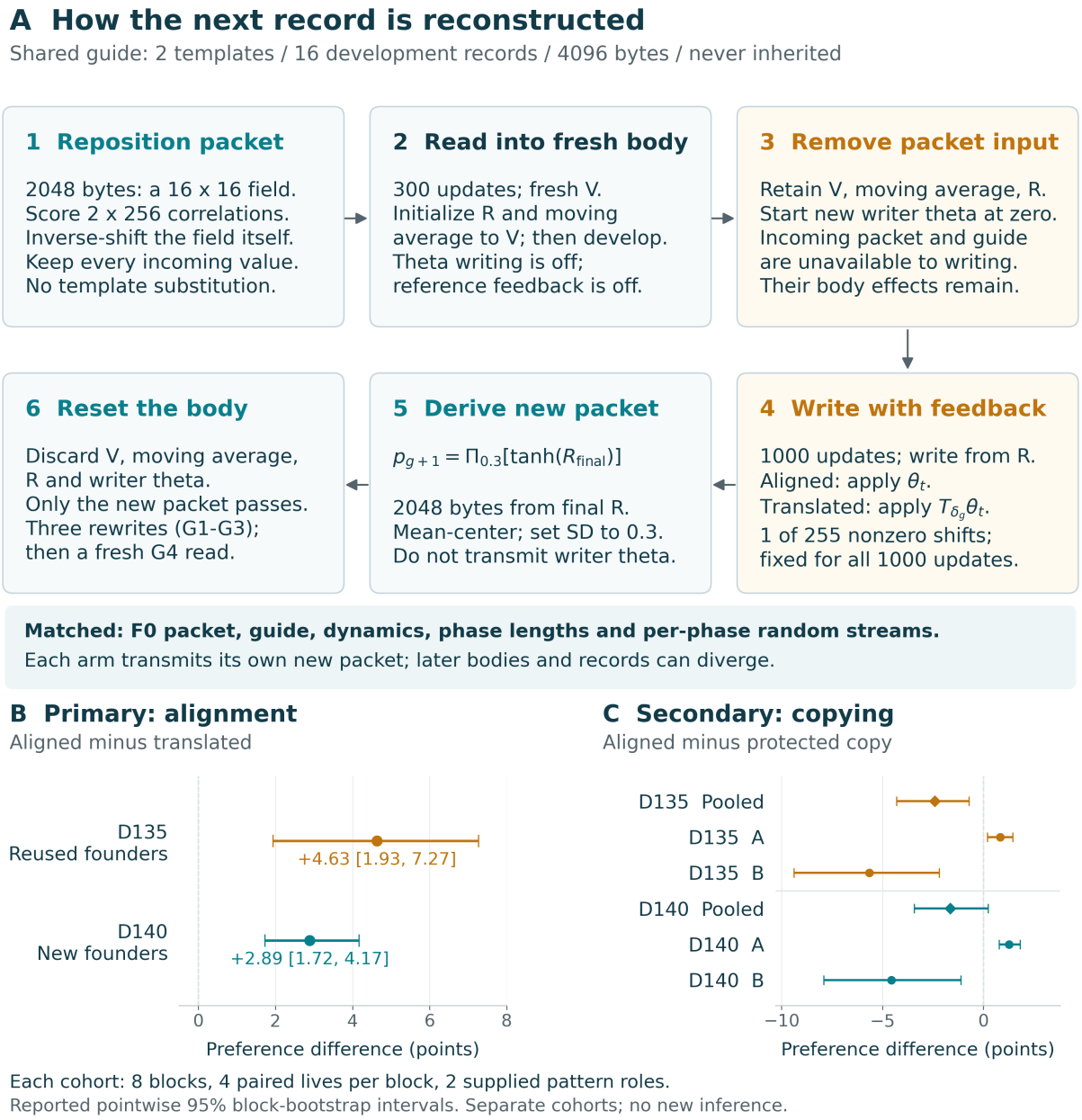



**Figure 8: Writer placement changes reconstruction; copying fidelity is a separate comparison.** A: guide-assisted repositioning, incoming-packet erasure before writing and fresh-body replacement. B: primary aligned-minus-translated effects in eight reused founder blocks and eight new blocks. C: secondary aligned-minus-protected-copy effects, pooled and by target role. B/C use qualified-preference differences and the reports' pointwise 95% whole-block bootstrap intervals. Each block contains four paired lives and two supplied roles; cohorts are not pooled. Panel A is a protocol schematic; B/C are measured contrasts. E29

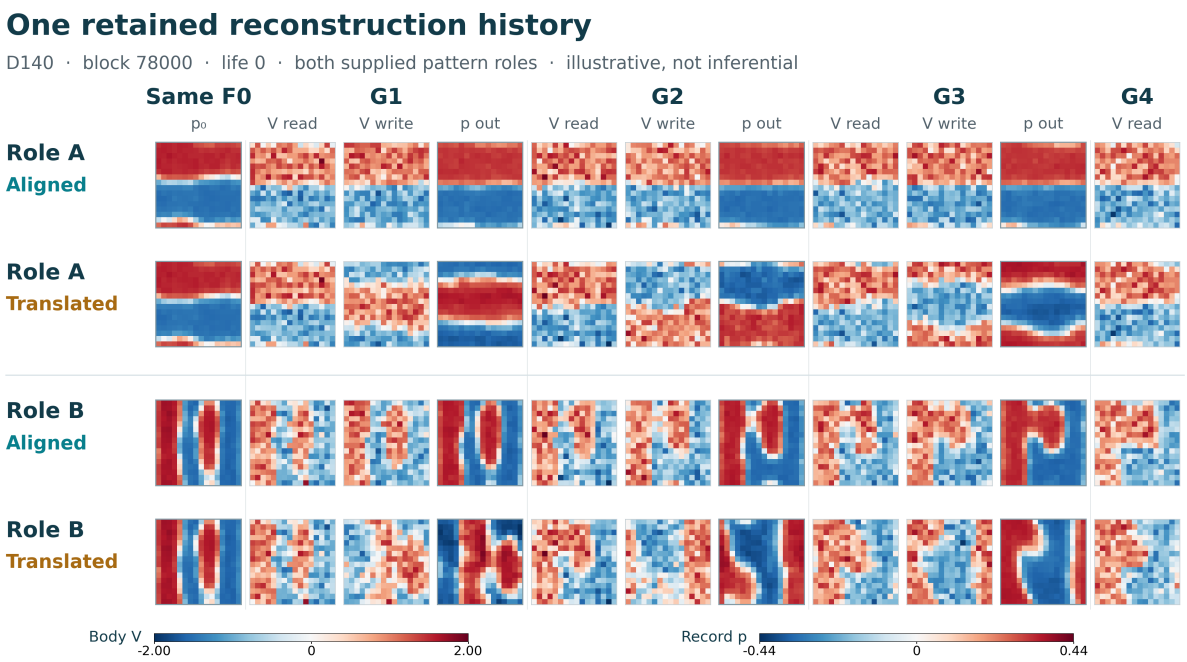



**Figure 9: One retained reconstruction history.** The smallest registered new-founder block (78000), first life, both target roles and writer arms. This identifier-based editorial rule preceded inspection of the selected fields but followed knowledge of aggregate results. The same initial record begins each paired chain. G1–G3 show activity after reading, activity after writing and the outgoing reference-derived record; G4 is a fresh terminal read. Only outgoing records cross boundaries. Tiles retain stored coordinates without display alignment or smoothing, with one shared body color scale and one record scale. This is an illustration, not an effect estimate or additional replication. Extraction and selection are documented in C.9. E29

Coordinate and shape fidelity answer different questions. The alignment effect is established on qualified preference; a separate analysis allowing all translations finds that within-family shape detail also declines along the aligned chain. It does not estimate the treatment contrast under a translation-invariant observer (SI C.8–C.9). E28; E29

Nor is identical placement always best. **Compatible feedback** (D145) uses shifts preserving the target family's symmetry. It exceeds zero-shift preference for pattern B by **8.11 points [6.07, 10.57]** over sixteen fresh blocks. A secondary comparison exceeds copying the imperfect original record by **3.83 points [2.21, 5.43]**. The shared guide and family-specific shifts supply prior structure; target-family improvement does not establish preservation of all founder detail. E40

In a separate **Hopfield reader confirmation** (H004), sustained stronger cue coupling supports **92.53% terminal function**, versus **43.14%** when supplied only at the final read, across 72 fresh networks and sixteen replacements. Terminal readers are matched, locating the difference in developed records. Protected copying remains stronger at **98.87%**. This supports a reader-history effect in another architecture, not a controlled ranking of architectures (SI C.41). E86

These are bounded examples of **active form**: embodied organization contributes causally to maintaining or rebuilding conditions for later expression. Each claim identifies the maintained distinction, information route and removing intervention. Simple feedback can qualify; neither

the term nor these experiments establishes autonomous maintenance or a nonphysical source of correction.

Recognizable history need not be usable for renewal. A retained cellular-automaton report finds **84.9% history-decoding accuracy**, versus **50%** for the sham, from a local patch after the incoming carrier is erased; none of twelve tested live writers passes renewal screening. The decoder and writers use different transformations and are not resource-matched competitors. The result distinguishes information available to the tested observer from information the tested writing routes can turn into a useful successor record (SI C.37). E81

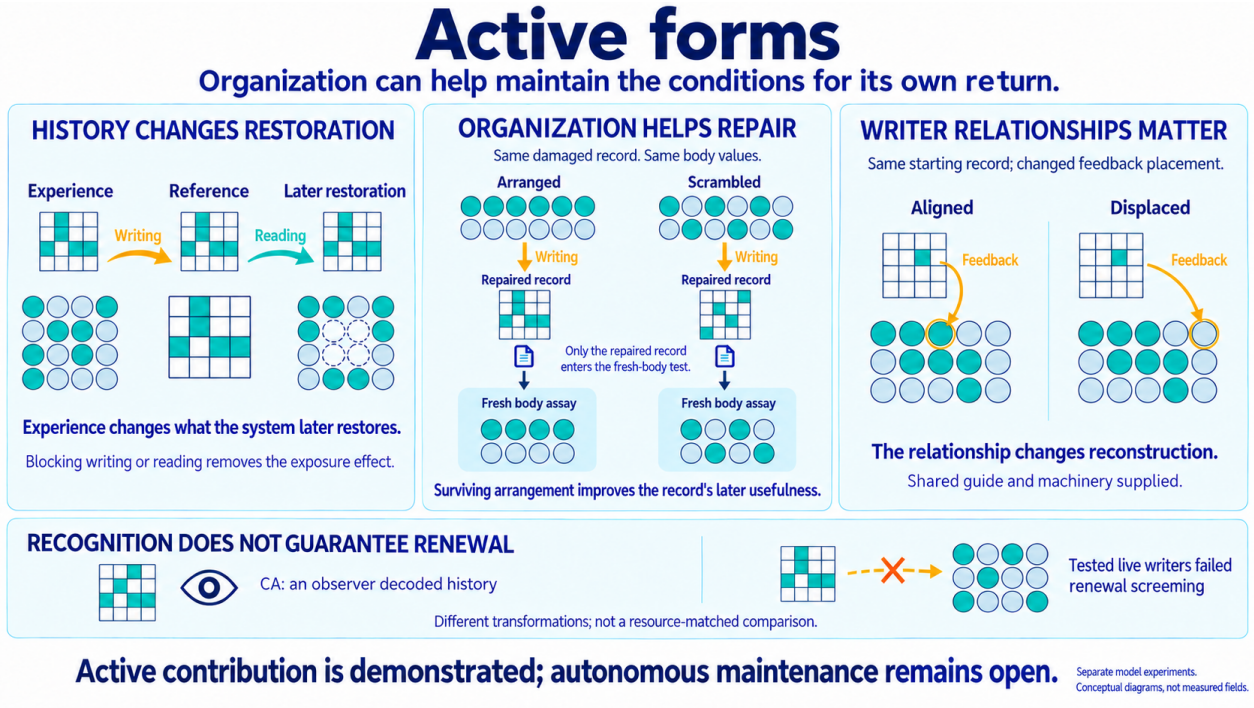



**Figure 10: Active forms: selected causal tests.** Separate experiments identify an activity-written reference, a contribution from surviving spatial organization to record repair, and dependence on writer placement. In the repair comparison, matched damaged records and body-value distributions isolate arrangement; only the outgoing record enters fresh-body assays. The spatial writer comparison changes feedback placement within supplied machinery and a shared guide. The CA strip contrasts external decoding with unsuccessful live-writer screening using different transformations, not matched computational resources. These conceptual diagrams illustrate active contributions; message-specific corrective content and autonomous maintenance remain unresolved. E08; E09; E10; E29; E81

## 5. What record structure supports functional reconstruction?

### 5.1. A taught response through damaged rebuilds

The **damaged functional-renewal confirmation** (E041) uses a 16 × 16 activity sheet with 512 reciprocal neighbor weights. Teaching clamps an input and two output patches; activity writes the weight record. With cue c and taught orientation s, the task is

$$(A, B) = (sc, -sc), \qquad c, s \in \{-1, +1\}.$$

| Cue | Rule I: required signs (A, B) | Rule II: required signs (A, B) |
|---|---|---|
| $-1$ | $(-, +)$ | $(+, -)$ |
| $+1$ | $(+, -)$ | $(-, +)$ |

Both cues must produce the required signs and output-patch means exceeding 0.25 in magnitude over the final 32 updates. During rebuilding, only input ports are clamped; outputs are not retaught. The local writer combines neighboring activities across both supplied cue trials, using their contrast product when exactly nonzero and ordinary coactivity otherwise. It receives no output label or direct incoming weights. The body is discarded; only newly written weights cross replacement. Thus spatial locality still depends on coordinated observations across two trials. E63

Across **32 fresh independent blocks, 128 founders and 512 lineages**, each body loses 24 non-port cells under a new mask. After 32 damaged rebuilds, full rewriting supports **83.59% terminal success [80.66, 86.33]** in an additional injured fresh recipient. Only **205/512 histories (40.04%)** pass every saved rebuilding-body readout; **307/512** fail at least once. Terminal reconstruction and uninterrupted performance are different endpoints (Figure 11). E63; E64

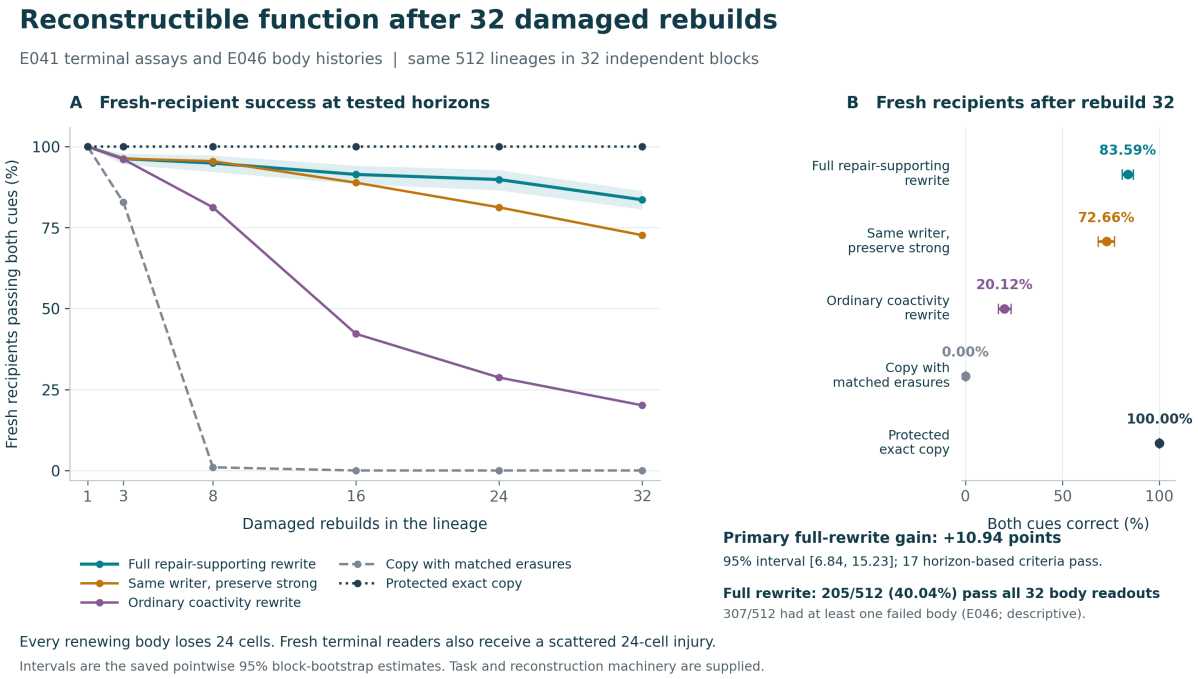



**Figure 11: Terminal reconstruction and uninterrupted performance are different endpoints.** One fresh cohort supplies all plotted policies and six measured post-acquisition horizons. A: injured terminal reconstruction at each horizon; the band belongs to full rewriting. B: horizon-32 estimates and saved pointwise 95% whole-block intervals. Full rewriting reaches 83.59%; only 205/512 of these histories pass every recorded rebuilding-body readout, while 307 have at least one failure. This history count is descriptive, not another cohort or terminal estimate. Protected copying avoids rebuilding-body losses; matched-erasure copying accumulates the corresponding edge losses. The primary contrast is full rewriting versus preservation of strong weights. Reading/writing ablations, omitted from the plot, each score zero at horizon 32. E63; E64

Full rewriting exceeds preserving strong incoming weights, **72.66%**, by **10.94 points [6.84, 15.23]**, with 95% whole-block intervals. Protected copying retains **100%**, matched-erasure copying gives **0%**, and ordinary-coactivity rewriting gives **20.12%**; reading and writing ablations each give zero. The result supports a particular reconstruction mechanism under specified loss, not rewriting in general. SI C.23 retains all controls and developmental failures. E63

### 5.2. Rebuilding conditions change the mechanism

**High success can arise in more than one noise regime, with different writing routes and preferred policies.** In a fresh 32-block cue-pairing test (E051), matched-noise terminal success is **89.45%**; changing cue-noise correlation to 0.999 yields **97.07%**. Under independent noise, returning connections acquire nonzero cue contrasts, and ordinary fallback changes from 100% to 0%. Original and pure-contrast writers both reach **96.68%**, while strong-weight preservation reaches **99.80%**, reversing its matched-noise ranking. Pure-contrast writing fails under matched noise. These findings belong to a separate cohort from E041. E69

The matched-noise route is numerically sensitive. Another fresh cohort (E052) scores **83.40%** at reference, **81.25%** with float32 computation and storage, and **8.01%** after rounding weights to a $2^{-16}$ grid. Float32 storage alone retains **83.01%**. Separately, replacing the writer's exact-zero test with a $10^{-4}$ tolerance reduces E051 success from **89.45% to 2.54%**. Storage, computation, quantization and thresholding are distinct interventions; small nonzero values matter to this writer and its normalized reader. E69; E70

The **joint numerical challenge** (E053) combines float32 computation, rounding, noise on nonzero stored weights and a positive normalization floor. Under independent rebuilding noise, **472/512 (92.19%)** succeed, versus **487/512 (95.12%)** at reference. The paired difference is **−2.93 points [−5.08, −1.17]**, across 32 new blocks. Success remains high, but the interval supports some loss and narrowly misses the five-point noninferiority allowance; the complete qualification fails (Figure 12). The matched-noise secondary contrast is **63.67% versus 82.23%**. E75

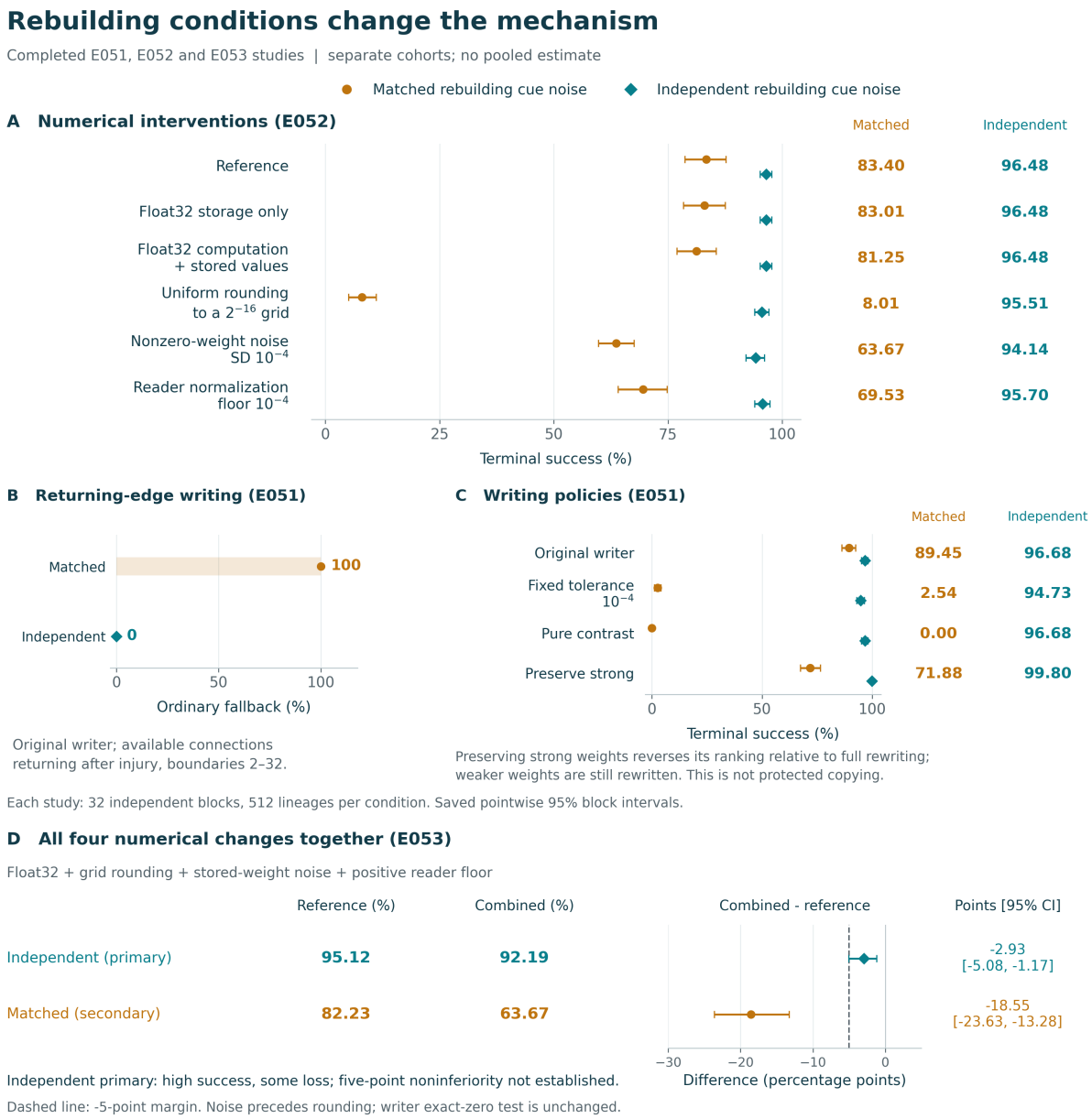



**Figure 12: Successful reconstruction can depend on different writing routes.** A: E052 reference, storage-only control and four prespecified numerical challenges applied separately, under matched or independent rebuilding cue noise. B: E051 original-writer fallback on available returning connections, summarized over boundaries 2–32. C: E051 terminal success for four writing policies. D: E053 reference and combined-condition success, with original paired-difference intervals; the dashed line marks the −5-point margin. Each study has its own 32 independent blocks and 512 lineages per condition; studies are not pooled. Success means both cues pass in an injured fresh recipient after 32 rebuilds. Error bars retain saved pointwise 95% whole-block intervals; B's intervals collapse at 100% and 0%. Strong preservation rewrites weaker weights and is not protected copying. E052's independent-noise comparisons remain secondary to its failed matched-noise qualification. E053 retains high independent-noise success with some loss, while its five-point noninferiority requirement remains unestablished. E69; E70; E75

The combined channel is not necessarily harder than each part. Noise precedes rounding and can push a tiny nonzero weight across a rounding threshold; original zeros remain protected. E053's matched-noise cohort scores **8.20%** with rounding alone and **63.48%** with noise then rounding. Its factorial comparisons support nonadditivity but do not isolate channel order: each condition develops its own history. Teaching, initial activity and terminal cue trials remain paired, and a finite writer tolerance is not among the four joint changes. SI C.25–C.26/C.31 preserves the qualifications. E75

### 5.3. Signs, magnitudes and timing

For ideal pure-contrast capture, $w_{ij} = d_i d_j$ with $d_i = (x_i^+ - x_i^-)/2$. Nonzero edge signs factor as $s_i s_j$, where $s_i = \text{sign}(d_i)$; cycle sign products are positive. Reorienting activity by s converts normalized coupling into a positive weighted average, canceling the receiving cell's absolute contrast magnitude. Structural balance is established signed-network mathematics, not a new theorem here. It also does not guarantee success: connectivity, ports, finite reading time and writing history remain consequential; fallback and preservation mixtures need not be balanced. SI C.23 gives the derivation. Altafini, 2013

**Record-feature interventions** (E054) probe all retained E051 records in fresh common readers without further writing. Permuting magnitudes within sign classes reduces matched-history success from **86.52% to 12.11%**, but independent-history success from **95.90% to 94.34%**. A post hoc whole-block paired interaction is **72.85 points [68.55, 76.95]**, a pointwise 95% interval over 32 blocks. Similar baseline success therefore relies differently on magnitude organization across these histories. E76

Incompatible signs are more damaging than removing the same edges: sign flips give **13.28%/16.02%** for matched/independent histories, versus **81.25%/94.92%** after deletion. Coordinated sign switching also transforms reset and noise while preserving ports; its unchanged outcome is a symmetry check, not robustness to arbitrary sign corruption. These are fresh-reader feature assays, distinct from E051's earlier terminal outcomes. SI C.32 retains policy interactions and adjustment. E76

A **capture-timing confirmation** (E057) shows that ongoing activity is not always ready to write. Under the primary matched-noise history, allowing 512 recovery updates after an activity pulse improves the pulse-versus-sham contrast by **2.34 points [0.59, 4.30]** across 32 fresh blocks. The interval has family-adjusted 98.75% coverage. Delayed capture needs no unstressed record, whereas a separate magnitude-substitution control receives one (SI C.39). E84

The copying comparisons set the limit. In the four-message reference chain, recognition falls from **100% at generation one to 27.3% at twelve**, near 25% chance, while protected copying retains 100%. A guide-free acquired-threshold renewal comparison remains unresolved. Reconstruction can counter specified loss or improve expression using supplied structure; it can also erase distinctions that copying preserves. These studies do not establish loss or prior structure as necessary or sufficient conditions for benefit, or compare equal total costs. E12; E27; E40

# 6. Can record structure improve forecasts or guide repair?

## 6.1. Frozen forecasts add information beyond expression

The **prospective continuation-forecast test** (TT008) freezes predictions before continuation on **64 fresh blocks containing 4,096 related records**. A signed-transfer diagnostic uses the known ports and taught orientation. Its Brier error is **0.1291**, versus **0.1378** for expression/history and **0.1382** when unsigned access is added; both primary improvement bounds exceed the prespecified margin. Lower error means more accurate probability forecasts. E71

| Frozen predictor | Information beyond shared task-aligned outputs and writer history | Record-derived operator applications |
|---|---|---|
| Expression/history | None | 0 |
| Unsigned-access comparator | Absolute normalized transfer at known ports | 64 |
| Signed-transfer candidate | Unsigned and signed, task-aligned transfer | 64 + 64 |

These diagnostics reclamp the input while iterating a linear operator. They are model-informed features, not the body's nonlinear continuation. Computational and probing budgets are not matched. E71

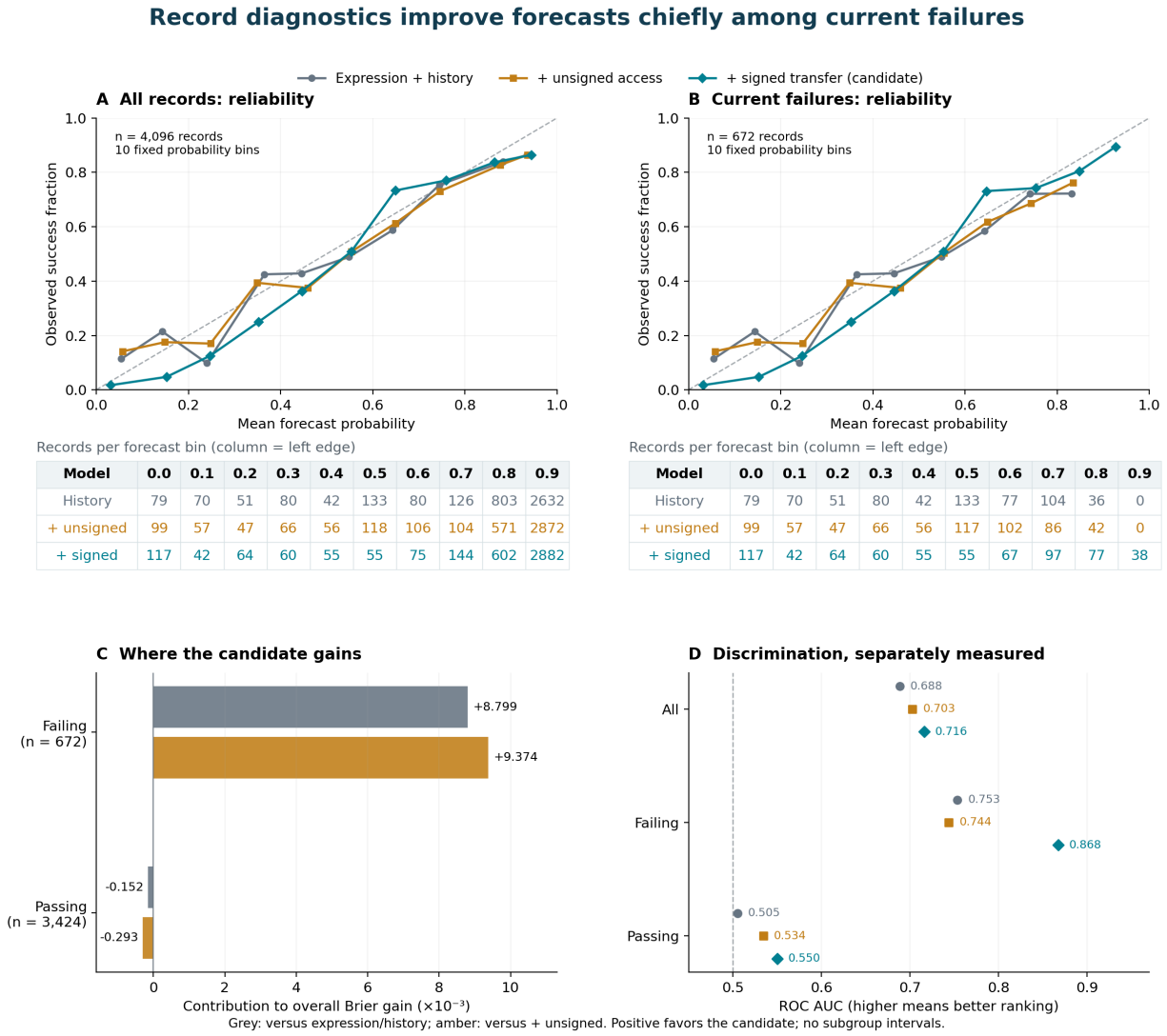



**Figure 13: Forecast improvement is concentrated among currently failing bodies.** A/B: reliability of the candidate and its two primary rivals across all records and the current-failure subgroup. Points show mean probability and observed success in ten fixed, equal-width bins; empty bins are omitted and connecting lines are visual guides. Tables give record counts for each model and bin, with column headers identifying left edges and the final bin including 1. Counts describe support, not additional independent units. C: each current-status subgroup's contribution to the overall Brier improvement, in thousandths of score units. The small unfavorable passing contribution remains visible. D: descriptive ROC AUC for all, failing and passing records, with half credit for tied ranks. The original 4,096 records belong to 64 independent blocks; plotted subgroup summaries have no new uncertainty intervals. Reliability and AUC are post hoc descriptions of frozen forecasts, with no fitting or recalibration. They do not partition the unbinned Brier score into causal or independent components. The separate decision task remains tied. E71

The gain concentrates among **672 currently failing bodies**: their contribution to overall Brier improvement is **+0.008799**, while current passes contribute **−0.000152**. Post hoc checks of the frozen forecasts show better ranking among failures, but slightly worse fixed-bin calibration error; all models overpredict on average. The separate decision task gives all primary predictors and always choosing spaced history the same **528 successes in 602 pairs**. Better probabilities do not automatically give better choices. SI C.27 retains inputs, margins and subgroup definitions. E71

## 6.2. Surviving structure guides a fixed repair

The **prospective surviving-record repair test** (TT014) freezes an algorithm before a fresh cohort. It infers cell signs by minimizing weighted disagreement with surviving connections, using symmetrized reader-normalized weights and positive input pins. It fills every exactly zero connection with the inferred sign product times the median surviving magnitude, preserving all nonzero weights. Disconnected unpinned components receive a fixed orientation convention. The full objective, starts, updates, ties and stopping rules remain inspectable in SI Algorithm 1, C.34. E78

Repair receives no previous packet, desired output label, body state or future outcome. Learned structure remains in the record; the lattice, pins, solver and reader are supplied. Across **64 fresh blocks**, all **1,024 outgoing generation-14 records** receive eight paired fresh injury/reset/noise assessments per condition. Repair raises success from **81.52% to 87.45%**, a gain of **5.93 points [4.88, 6.99]**, using the original two-sided stratified interval. Shuffling only the added signs gives **25.83%**. Registered one-sided lower bounds clear both practical margins. E78

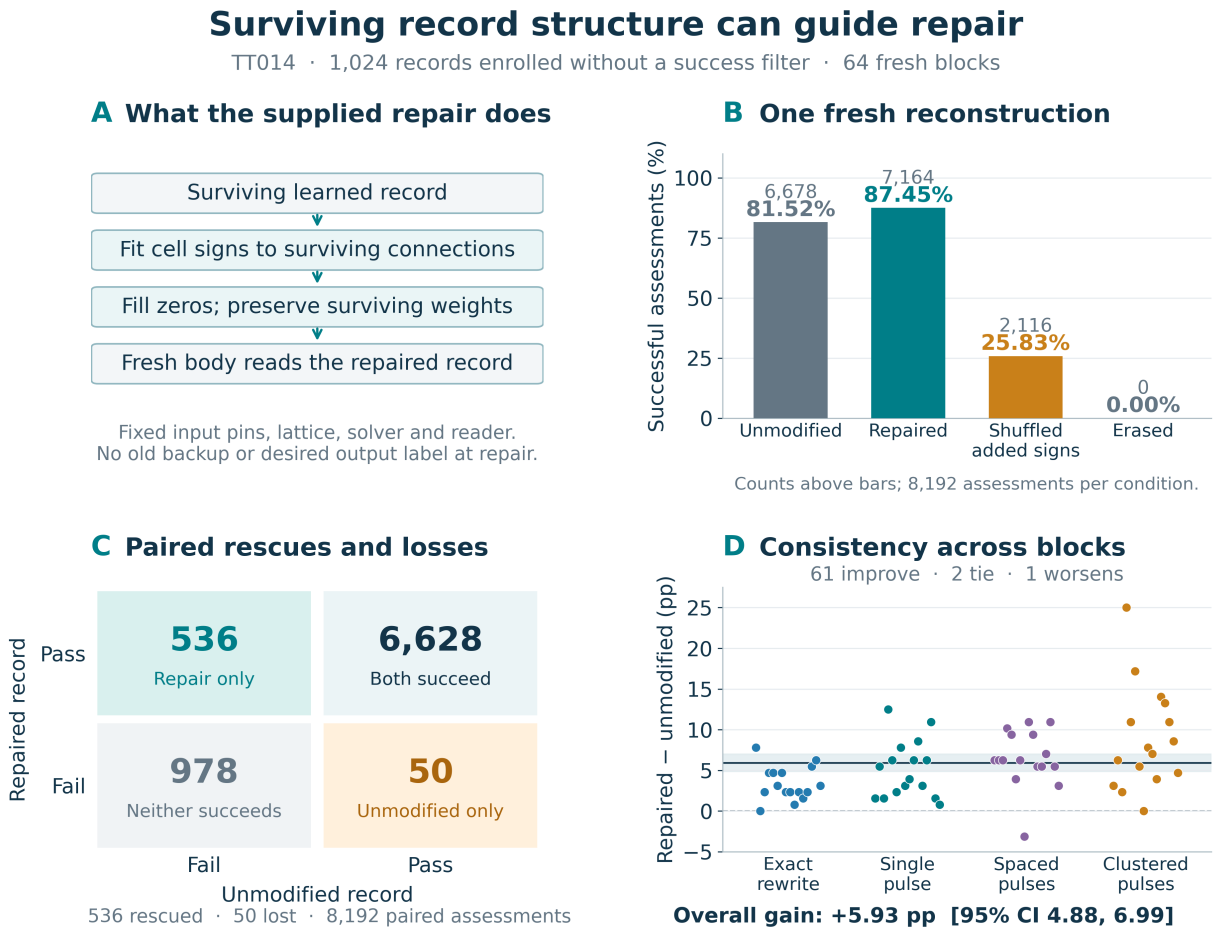



**Figure 14: Surviving learned structure guides a useful repair.** A: schematic operation of the frozen surviving-record procedure; no saved field is depicted. B: original success rates and counts for all four conditions, with eight paired fresh injury/reset/noise assessments for each of 1,024 records. C: all 8,192 paired repaired-versus-unmodified outcomes, including 536 rescues and 50 losses. D: all 64 independent block differences, grouped by the four acquisition histories; horizontal offsets follow block order. The solid line and shaded band show the original overall mean and two-sided 95% stratified Welch–Satterthwaite interval, not uncertainty for individual blocks or histories. Independent blocks, not repeated assessments, are the inference unit. The repair uses no desired output label or previous packet; the input pins, lattice, solver and reader remain supplied. Success concerns one fresh reconstruction under both cues. TT016's subsequent serial test is separate. E78

Improvements occur in **61 of 64 blocks**, with two ties and one deterioration. Among 8,192 paired assessments, **536 succeed only after repair and 50 only without it**. Blocks are the independent inference units. The practical gain over an unmodified record and the larger shuffled-placement contrast answer separate questions; this qualification concerns one fresh reconstruction. E78

### 6.3. Repeated repair helps, but choosing when remains open

A separate **prospective serial-repair confirmation** (TT016) enrolls all 1,024 records in 64 fresh blocks, including initial failures, and follows eighteen additional rebuilds. Each branch transmits its own successor. Under independent ongoing cue noise, repair at every rebuild raises **uninterrupted success from 77.54% to 89.84%**, or **12.30 points [9.71, 14.90]**. The registered one-sided bound clears the five-point margin. E83

| Repair policy | All eighteen bodies pass | Last rebuilding body passes |
|---|---|---|
| No repair | 794/1,024 (77.54%) | 880/1,024 (85.94%) |
| First rebuild only | 856/1,024 (83.59%) | 912/1,024 (89.06%) |
| Every rebuild | 920/1,024 (89.84%) | 934/1,024 (91.21%) |
| Shuffled added signs every rebuild | 0/1,024 | 87/1,024 (8.50%) |

These are rebuilding-body endpoints, not E041's additional injured terminal recipient. Repeated supplied correction improves continuity, at greater computation; shuffled repair can itself harm. This is not autonomous maintenance. SI C.38 separates the serial cohort from the one-read test. E83

Policy tests preserve a useful failure. A frozen warning policy with nine repair calls gives **85.55%** uninterrupted success versus **86.33%** for periodic repair across sixteen fresh blocks. A separate 132-block prospective study supports benefit from eighteen versus one repair but leaves the stronger efficiency criterion unresolved. Calls do not equal total computational cost. A useful operation and an informative predictor therefore do not yet supply a superior intervention policy. E88

## 7. What would a predictive theory of continued reconstruction require?

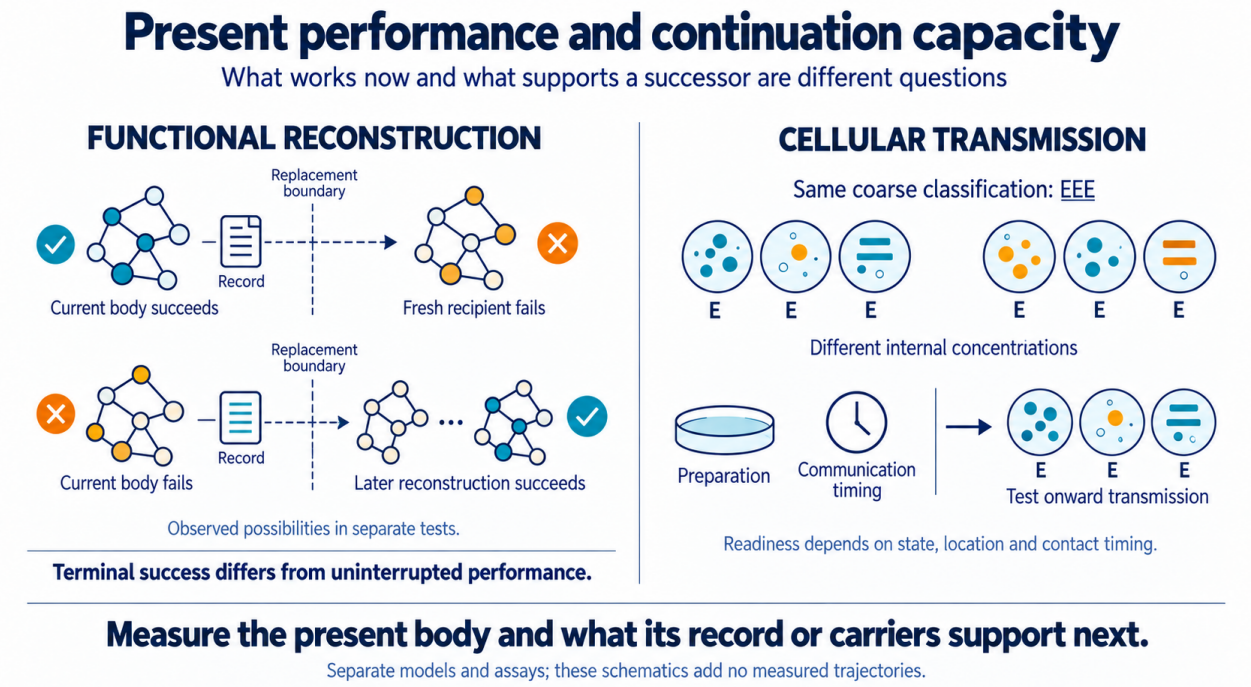



**Figure 15: Present performance and continuation capacity.** Left: separate functional tests show successful bodies followed by unsuccessful recipients, and failed body readouts followed by later recovery. Receiving conditions also affect these outcomes. Right: cellular trios can share an EEE classification while differing in concentrations and transmission readiness. Preparation and communication timing have context-dependent effects. The diagrams summarize observed possibilities across separate assays, without supplying new trajectories or success rates. E46; E61; E64; E65; E67

### 7.1. Capacity depends on the future and the recipient

A matched **recipient-reader intervention** (E023) makes the apparatus dependence explicit. Normalizing incoming drive to surviving connection strength, while holding inherited records and recipient conditions matched, raises injured-recipient success from **27.5% to 99.6%**, a gain of **72.07 points [68.16, 75.98]** with a 95% whole-block interval across 32 independent blocks. The same record can support different outcomes with a different reader; this is not additional transmitted information (SI C.23). E61

Saved histories show why present expression is insufficient. Of E041's **307 ever-failed histories, 236** succeed at the additional terminal assay; **192 of 205** always-passing histories succeed there. Separately, 243/281 first-failure histories with enough follow-up recover within three subsequent bodies. These are descriptive counts, not repeated estimates of each record's conditional capacity. Injury and birth conditions vary across generations. E64

Repeated fresh readings sharpen the distinction. For 64 selected failed bodies from eight exposed blocks, outgoing-minus-incoming success is **+7.03 points [1.03, 13.48]** under the original injury, but **−6.13 [−14.68, 2.24]** under fresh injuries. The fresh-injury benefit prediction fails. Con-

versely, another saved-state intervention repairs all 46 selected later failures under exactly their original injury, reset and noise, isolating immediate rescue from a luckier draw. Neither assay follows repeated futures through successive rewrites. SI C.27/C.44 retains selection and inference limits. E72; E89

Cellular readiness provides a complementary causal example. Trios with the same coarse EEE label differ after preparation; transferring either the prepared newcomer's complete state or the prepared surviving neighbor's state rescues transmission, while leaving both unprepared fails. RNA and protein substitutions have different sufficiency at the two locations and across backgrounds. Ordinary and written origins agree. Specific internal differences matter beyond classification, without identifying a unique memory molecule. E65; E66

A completed **communication-timing panel** (P32) succeeds across all sixteen related history/sequence/source/resolution variants with a five-unit immediate RNA-transport pause, the shortest tested. Continuous communication and blocking the entire challenge fail. Delayed pause starts also fail beyond the tested window. Final retention and onward transmission coexist with temporary label loss; numerical resolutions and related histories are model panels, not independent population replicates. SI C.24/C.40 gives the complete schedule and controls. E85

The source-model mapping remains bounded: the 17-entry BITSEY state includes primary concentrations, fixed inputs and derived RNA entries; E/M is a separate observer. Added contact schedules and substitutions differ from native equations, and the integration procedure was changed. Historical source-output checks are reported, but exact correspondence to source-paper table geometries and labels remains unresolved (SI C.7). This documentation boundary accompanies the cellular claims.

### 7.2. Reconstructive Dynamics Theory as a candidate

**Reconstructive Dynamics Theory** (**RDT**) takes the complete cycle as its object: record → development → body → writing or repair → successor record. Its conjecture is that correction of disruptive variation and preservation of acquired distinctions can jointly predict continuation under declared disturbances. Converging on a common prototype can restore an appearance while erasing historical differences; retaining differences can still fail to build a functioning successor.

Candidate measurements concern correction of perturbations, proximity to loss of the acquired distinction, and the disturbance introduced at each renewal.

A useful theory must predict untouched continuations or interventions beyond strong comparators with comparable information and resources. Persistent reader or writer state must be counted in its conditioning variables. Section 2 defines what to estimate; it does not supply this law. TT008 and the repair confirmations are within-apparatus advances, compatible with conventional prediction and error correction.

Failures help specify the gap. A prospective tissue forecast leaves its challenge-versus-passive primary comparison unresolved. An earlier short-to-long forecast permits almost the full range of long-term success under its confidence construction. Finite decision models adapt better to broadened change schedules while still failing economical-recovery criteria. These constrain the tested candidates rather than proving that compact prediction is impossible. Full forecasting and FEP development remain in the repository's extended evidence, with methods in SI C.15/C.19–C.22/C.36/C.43. E55; E80; E88

The next standard is prospective prediction across changed record histories, damage distributions and reader–writer conditions. Repeated continuations from exactly restored states would distinguish reliable record-conditioned capacity from favorable later circumstances. The theory and theory-tests lanes formulate and challenge candidates within the same research program; ongoing computations provide no assumed results here.

## 8. Discussion: embodied access to form

### 8.1. Biological relevance of the reconstruction boundary

Feedback memory has substantial biological and modeling precedents. Dodd's nucleosome model and Owen's contact-dependent reader–writer model explain restoration under erosion through explicit interactions. Chromatin-regulator exposure, substrate stiffness and transient planarian bioelectric interventions can alter later responses. Existing tissue also participates in rebuilding: planarian muscle expresses positional signals, including injury responses without the stem cells making replacement tissue. These comparisons motivate information routes to test; they do not validate identities between our fields and biological stores. Dodd et al., 2007; Owen et al., 2023; Bintu et al., 2016; Yang et al., 2014; Durant et al., 2017; Witchley et al., 2013

The transferable contribution is an experimental distinction: identify retained state, remove the continuing cue, intervene on its reader or writer, and test the appropriate later boundary. Within-body persistence, a successful repair and transmission into replacements require different evidence. For reciprocal repair, identifying message-specific corrective content remains the next target; usefulness after repair alone leaves its source unresolved.

### 8.2. Where forms come from: possibility and historical access

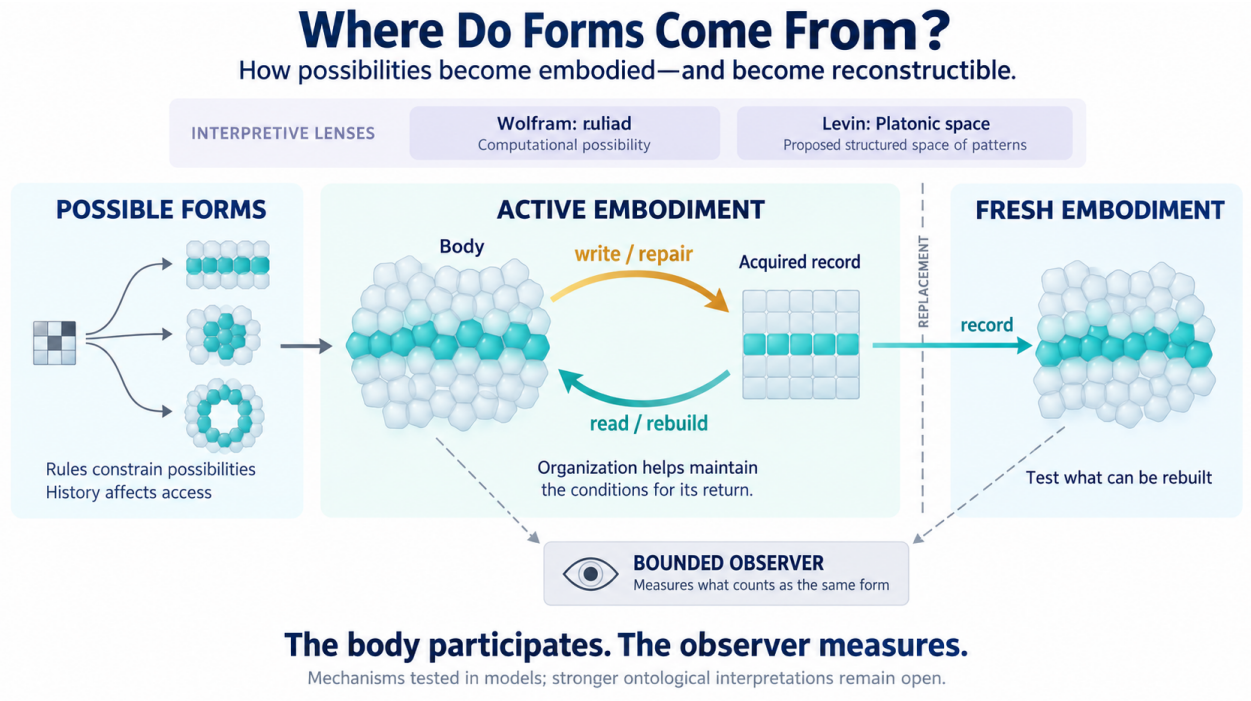



**Figure 16: Where do forms come from?** Rules constrain possibilities; histories and embodied reading–writing relationships influence which forms become reconstructible. The central loop synthesizes acquired-reference, reciprocal-repair and replacement tests across separate experiments; it does not depict one autonomously maintained apparatus. Only the declared record crosses the illustrated replacement boundary, with recipient machinery supplied. The bounded observer measures distinctions separately from the causal reader and writer. Wolfram's ruliad and Levin's proposed Platonic space provide interpretive lenses whose stronger ontological claims remain open. Cell patterns are conceptual illustrations, not saved simulation fields.

The origin of an instance and the acquisition of reliable access to it are different problems. Supplied dynamics and embodiment constrain possibilities; a history realizes an organization; acquired records and working readers and writers can make its return depend on that history. Unprescribed capture connects occurrence to historical specificity while leaving the origin of the general machinery explicit.

Wolfram's ruliad and bounded-observer account motivate distinctions among permitted dynamics, accessible experience and recognition. A record can change a fast subsystem's destinations or their accessibility while the encompassing body-plus-record law stays fixed. Our finite maps and observation contrasts translate these questions into bounded tests, not representative samples of the entire ruliad. E33; Wolfram, 2022; Wolfram, 2023

Levin's proposed Platonic space emphasizes embodiment as an interface to structured possibilities, potentially including active patterns. The operational question is whether that relationship predicts a new outcome or useful intervention. Our body–record and writer tests establish an active contribution from embodied organization: a body participates in making a form reconstructible, rather than serving only as a location where it appears. They do not distinguish independently

active nonphysical patterns from implemented dynamics; the stronger ontology remains open. Figures 10 and 16 retain the conceptual synthesis. Levin, 2026

An intervention-based atlas would have to predict held-out challenges, not merely group similar appearances. Computational mechanics and causal abstraction provide nearby precedents. Approximate similarity need not be transitive, and failure to resolve a difference does not establish equivalence. Crutchfield and Young, 1989; Rubenstein et al., 2017; Beckers and Halpern, 2019

### 8.3. From transmitted state to inherited competence

The largest missing link is an unprescribed useful competence that becomes durably inherited. Spatial capture retains modest historical specificity; the functional model renews a taught two-rule response; the cellular model transmits editable regulatory states. These are different achievements in different apparatuses. None combines them or establishes flexible problem-solving through complete machinery turnover.

A stronger test would make learned reader calibration, writer alignment or member roles vulnerable, then ask whether descendants restore them and compensate for an unfamiliar disruption. Levin's TAME framework motivates such functional tests across embodiments; collective chemotaxis and biofilm signaling demonstrate why intercellular relationships matter without themselves establishing inherited adaptability. Levin, 2022; Fields and Levin, 2022; Theveneau et al., 2010; Prindle et al., 2015

Supplied readers, resources and schedules remain outside what these systems maintain autonomously. Fidelity is not already viability, and an acquired reference does not establish autonomous valuation. Organizational closure and agency therefore pose additional requirements beyond the reconstruction effects reported here. Montévil and Mossio, 2015; Barandiaran et al., 2009; Kolchinsky and Wolpert, 2018

## 9. Conclusion

Acquired history, present expression and successor usefulness are experimentally separable. Surviving organization, reader–writer relationships and record features can change what a replacement reconstructs; a fixed repair procedure can improve both fresh reconstruction and serial maintenance. Protected-copy advantages, numerical sensitivities and failed policy predictions delimit these achievements. A predictive theory must explain those successes and failures under declared boundaries, rather than treating every returning pattern as the same phenomenon.

## Supporting information and availability

Open Supporting Information v1.1 and its methods guide. The guide gives SI page numbers for the principal methods; main-text SI citations link to the corresponding PDF pages. Cross-document page jumps depend on the PDF viewer.

This v1.1 revision reorganizes the evidence selected through 12 October 2026 without rerunning simulations, fitting models or changing campaign qualifications. The separate **Supporting Information v1.1** retains all 44 methods sections, the complete repair algorithm, inference

and source-model specifications, and five supporting figures. The public companion is Eidosoma/HowFormsBecomeReconstructible; the versioned paper package is distributed under paper/v1.1, with PDF hashes identifying the exact files.

The complete evidence ledger, extended comparisons, code, compact data and provenance remain in the repository. E-number citations refer to its stable evidence identifiers. Earlier figure/section numbers in those historical documents refer to the full v1.0 layout; the versioned package supplies a correspondence table. The availability guide identifies omitted raw arrays and external-model dependencies. Source hashes and packaging checks do not establish complete campaign replay. Post hoc E014, E054 and TT008 analyses retain that status; saved histories remain illustrations. The public companion's first review snapshot is commit 036bd902.

## Human direction, AI assistance and registration

Robert Bjarnason conceived and directs the program and is the accountable author. Eidosoma AI Scientist, Codex Sol and Claude Fable assisted implementation, simulation, analysis and writing as documented by the companion; Codex assisted this synthesis and reorganization. Conceptual images used an image-generation tool. AI systems are tools, not authors.

Separate implementations belong to one human-directed program, not independent laboratories. Protocol freezes are campaign-local, not independently timestamped public registrations. Outcome-informed follow-ups, related panels and fresh confirmations remain distinct. Reviewer identities and independence are not asserted. This is preparation for the first Zenodo release; no deposit or DOI is claimed.

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